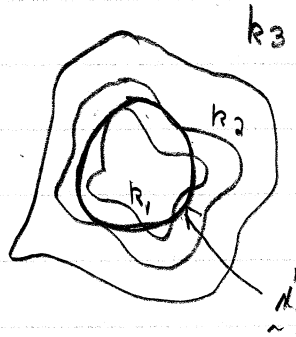
STATISTICS 652

LECTURE 9 - 2026

REVIEW: $\sup_{\underset{\sim}{x} \in \mathbb{R}^q,\ \underset{\sim}{x}'\underset{\sim}{x}=1} g(\underset{\sim}{x})$



CONCENTRIC ORBITS / CONTOURS FOR $g(\underset{\sim}{x}) = k$

FOR RAYLEIGH QUOTIENTS

$$\sup_{\underset{\sim}{x} \in \mathbb{R}^q} \frac{\underset{\sim}{x}'A\underset{\sim}{x}}{\underset{\sim}{x}'\underset{\sim}{x}} = \sup_{\underset{\sim}{x} \in \mathbb{R}^q,\ \underset{\sim}{x}'\underset{\sim}{x}=1} g(\underset{\sim}{x}) \quad \text{where } g(\underset{\sim}{x}) = \underset{\sim}{x}'A\underset{\sim}{x}$$

APPLICATION TO PRINCIPAL COMPONENTS: $\underset{\sim}{X} \in \mathbb{R}^q$, $\Sigma = \mathrm{VAR}(\underset{\sim}{X}) \geq 0$

SVD: $\Sigma = P\Delta P'$, $P = [\underset{\sim}{p}_1 \cdots \underset{\sim}{p}_q] \in O(q)$, $\Delta = \mathrm{diag}(\lambda_1, \ldots, \lambda_q)$, $\lambda_1 \geq \cdots \geq \lambda_q \geq 0$

1) $\sup_{\|\underset{\sim}{p}\|=1} \mathrm{VAR}(\underset{\sim}{p}'\underset{\sim}{X}) = \lambda_1$, obtained at $\underset{\sim}{p} = \underset{\sim}{p}_1$

2) $\sup_{\|\underset{\sim}{p}\|=1} \mathrm{VAR}(\underset{\sim}{p}'\underset{\sim}{X}) = \lambda_2$ obtained at $\underset{\sim}{p} = \underset{\sim}{p}_2$

$\mathrm{cov}(\underset{\sim}{p}_1'\underset{\sim}{X}, \underset{\sim}{p}_1'\underset{\sim}{X}) = 0$ ⋮

k) $\sup_{\|\underset{\sim}{p}\|=1} \mathrm{VAR}(\underset{\sim}{p}'\underset{\sim}{X}) = \lambda_k$ obtained at $\underset{\sim}{p} = \underset{\sim}{p}_k$

$\mathrm{cov}(\underset{\sim}{p}_j'\underset{\sim}{X}, \underset{\sim}{p}_j'\underset{\sim}{X}) = 0$ ⋮

$j = 1, \ldots, k-1$ ETC

NOTE: $\mathrm{Cov}(\underset{\sim}{p}_j'\underset{\sim}{X}, \underset{\sim}{p}_j'\underset{\sim}{X}) = 0 \iff 0 = \underset{\sim}{p}_j'\Sigma\underset{\sim}{p}_j = \lambda_j \underset{\sim}{p}_j'\underset{\sim}{p}_j \iff \underset{\sim}{p}_j'\underset{\sim}{p}_j = 0$.

$\mathrm{VAR}(\underset{\sim}{p}'\underset{\sim}{X}) = \underset{\sim}{p}'\Sigma\underset{\sim}{p}$.

OTHER TRANSFORMATIONS : MULTIVARIATE STANDARDIZATION

UNIVARIATE : $X \in \mathbb{R}$, $Z = \frac{X-\mu}{\sigma} \Rightarrow E(Z) = 0$, $VAR(Z) = 1$

MULTIVARIATE : $\underset{\sim}{X} \in \mathbb{R}^q$, $E(\underset{\sim}{X}) = \underset{\sim}{\mu}_X$, $VAR(\underset{\sim}{X}) = \underset{\sim}{\Sigma}_X > 0$ Let $\underset{\sim}{\Sigma}_X = BB'$

$$\underset{\sim}{Z} = B^{-1}(\underset{\sim}{X} - \underset{\sim}{\mu}_X) \Rightarrow E[\underset{\sim}{Z}] = 0 \text{ AND } VAR(Z) = I_q$$

NOTE: $\|\underset{\sim}{Z}\| = \sqrt{\underset{\sim}{Z}'\underset{\sim}{Z}} = \sqrt{(X-\mu_X)'\underset{\sim}{\Sigma}_X^{-1}(X-\mu_X)} = D_q(\underset{\sim}{X})$ (VERIFY)

SPECIAL SQUARE ROOTS

i) UNIQUE SYMMETRIC POSITIVE DEFINITE SQUARE ROOT

$\underset{\sim}{\Sigma}^{1/2} = \underset{\sim}{P}\Delta^{1/2}\underset{\sim}{P}'$ where $\underset{\sim}{\Sigma} = \underset{\sim}{P}\Delta\underset{\sim}{P}'$ (NOT UNIQUE SYMMETRIC, WHY?)

ii) UNIQUE LOWER TRIANGULAR SQUARE ROOT (CHOLESKY DECOMPOSITION)

$\underset{\sim}{\Sigma} = T_L T_L'$ with T_L being lower trangular with non-negative diagonals

PROPERTIES OF SVD : $A_{q\times q}$ symmetric $A = P\Delta P'$

i) $A^2 = P\Delta^2 P'$ and $A^k = P\Delta^k P'$ $k = 1, 2, \ldots$ (VERIFY)

ii) FOR NONSINGULAR A, $A^{-1} = P\Delta^{-1}P'$ AND $A^k = P\Delta^k P'$ for $k = \pm 1, \pm 2, \ldots$

iii) FOR $A \geq 0$, $A^{1/2} = P\Delta^{1/2}P'$

DEFINE : iv) $A^0 = P\Delta^0 P' = I_q$

v) $A^\alpha = P\Delta^\alpha P'$ for $\alpha \geq 0$, $A \geq 0$ (why)

for $-\infty < \alpha < \infty$, $A > 0$.

NOTE: $TR(AB) = TR(BA)$, SO SYMMETRIC A, $TR(A) = TR(\Delta) = \sum_{j=1}^{q} \lambda_j$

MORE GENERALLY:

$h: \mathbb{R} \to \mathbb{R}$ CAN BE EXTENDED TO $h: \mathbb{R}_{SYM}^{q \times q} \to \mathbb{R}_{SYM}^{q \times q}$

where $\mathbb{R}_{SYM}^{q \times q}$ is space of SYMMETRIC MATRICES OF ORDER q

by definition: $h(A) = P h(\Delta) P'$ where $h(\Delta) = \text{diag}\left(h(\lambda_1), \ldots, h(\lambda_q)\right)$.

HW: i) SHOW $e^A = \sum_{k=0}^{\infty} A^k / k!$

ii) SHOW $(I-A)^{-1} = \sum_{k=0}^{\infty} A^k$ (Neumann expansion)

PROVIDED $-1 < \lambda_p \leq \lambda_1 < 1$.

iii) FIND A CLOSED FORM FOR $\sum_{k=0}^{m} A^k$ (NOTE: A NEED NOT BE SYMMETRIC)

iv) SHOW $h(A)$ IS WELL DEFINED.

COMMENT; THE SVD IS NOT WELL DEFINED EVEN THOUGH A IS WELL DEFINED,

i.e. $A = P \Delta P' = \sum_{j=1}^{q} \lambda_j \underset{\sim j}{P} \underset{\sim j}{P'}$ although $P = [\underset{\sim 1}{P} \cdots \underset{\sim q}{P}]$ IS NOT UNIQUELY DEFINED.

ALTERNATIVE REPRESENTATION FOR SPECTRAL VALUE DECOMPOSITION

(SEE PROBLEM 3.15, NLLL)

FOR $A \in \mathbb{R}_{SYM}^{q \times q}$

- LET $\lambda_{(1)} > \lambda_{(2)} > \cdots > \lambda_{(r)}$ denote its distinct roots

with $m_1, m_2, \ldots, m_r$ denoting their multiplicities, hence

$m_1 + m_2 + \cdots + m_r = q$.

- Denote the eigenspaces $\mathcal{E}(\lambda_{(k)}) = \{ \underset{\sim}{a} \mid A\underset{\sim}{a} = \lambda_{(k)} \underset{\sim}{a} \}$ $k = 1, \ldots, r$ which are orthogonal to each other, i.e. $\mathcal{E}(\lambda_{(k)}) \perp \mathcal{E}(\lambda_{(j)})$ $k \neq j$ and $\text{SPAN}(\mathcal{E}(\lambda_{(1)}), \ldots, \mathcal{E}(\lambda_{(r)})) = \mathbb{R}^q$, with $\dim(\mathcal{E}(\lambda_{(k)})) = m_k$.

- Let P_k denote the orthogonal projection operator (matrix) onto the subspace $\mathcal{E}(\lambda_{(k)})$, then $P_k = \sum_{j=J_k+1}^{J_k+m_k} \underset{\sim}{P}_j \underset{\sim}{P}_j' = \underset{\sim}{P}_k \underset{\sim}{P}_k'$ where $\underset{\sim}{P}_k = [\underset{\sim}{P}_{J_k+1}, \ldots, \underset{\sim}{P}_{J_k+m_k}]$, with $J_k = m_1 + \cdots + m_{k-1}$ being an orthonormal basis for $\mathcal{E}(\lambda_{(k)})$.

- Then $A = \sum_{k=1}^{r} \lambda_{(k)} P_k$ is a unique representation of the spectral value decomposition of A.

(Note: Although the basis $\underset{\sim}{P}_k$ is not uniquely defined, the eigenspace $\mathcal{E}(\lambda_{(k)})$ and orthogonal projection P_k are uniquely defined.)

General Basis: If $\underset{\sim}{X} = [\underset{\sim}{x}_1, \ldots, \underset{\sim}{x}_m]$ is any basis for a subspace $\mathcal{S} \subset \mathbb{R}^q$, with $\dim(\mathcal{S}) = m$, then the orthogonal projection onto $\mathcal{S}$ has the representation

$$P_{\mathcal{S}} = \underset{\sim}{X}(\underset{\sim}{X}'\underset{\sim}{X})^{-1}\underset{\sim}{X}' = \underset{\sim}{P}\underset{\sim}{P}' \quad \text{where } \underset{\sim}{P} = \underset{\sim}{X}(\underset{\sim}{X}'\underset{\sim}{X})^{-1/2}$$ which is an orthonormal basis.

OTHER MATRIX DECOMPOSITIONS

- **QR-DECOMPOSITION:** $M_{p\times r}$ with $p \geq r$, THEN THERE EXISTS $Q_{p\times r}$ with $Q'Q = I_r$ and $T_u \in \mathcal{T}_r^+$, the set of upper triangular $r\times r$ matrices with non-negative diagonals, such that $M = QT_u$

  PROOF: GRAM-SCHMIDT

- FOR $A_{q\times q} \geq 0$, $A = T_L T_L'$ with $T_L' \in \mathcal{T}_q^+$. CHOLESKY DECOMPOSITION

  PROOF: $A = MM' = T_u' Q' Q T_u = T_L T_L'$.

NOTE: T_L IS UNIQUE SINCE FOR $P \in \mathcal{O}(q)$, $PT_u \in \mathcal{T}_q^+$ IFF $P = I_q$.

- SINGULAR VALUE DECOMPOSITION

  FOR $M_{p\times r}$ with $p \geq r$, $M = Q\begin{pmatrix}\Delta\\ 0\end{pmatrix}H'$

  where $Q \in \mathcal{O}(p)$, $H \in \mathcal{O}(r)$ and $\Delta = \mathrm{diag}(\delta_1, \ldots, \delta_r)$ with $\delta_1 \geq \cdots \geq \delta_r \geq 0$.

NOTE: FOR $p \leq r$, APPLY TO M', i.e.

$M = Q(\Delta\ 0)H'$ with $\Delta = \mathrm{diag}(\delta_1, \ldots, \delta_p)$ with $\delta_1 \geq \cdots \geq \delta_p \geq 0$.

ALTERNATIVE FORM: $M = \sum_{j=1}^{\min(p,r)} \delta_j q_j h_j'$ $\quad Q = [q_1, \ldots, q_p]$, $H = [h_1, \ldots, h_p]$

SINGULAR VALUES: $\delta_1 \geq \cdots \geq \delta_{\min(p,r)} \geq 0$.

PROOF: Let $A = MM'$ and so A is $p\times p$ with $\mathrm{rank}(A) = r \leq p$.

$= Q\begin{pmatrix}\Lambda & 0\\ 0 & 0\end{pmatrix}Q'$, SVD, $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_r)$ $\lambda_1 \geq \cdots \geq \lambda_r$

$= Q\begin{pmatrix}\Delta\\ 0\end{pmatrix}(\Delta\ 0)Q'$ $\quad \Delta = \Lambda^{1/2}$

$\Rightarrow M = Q\begin{pmatrix}\Delta\\ 0\end{pmatrix}H'$ for some $H \in \mathcal{O}(r)$

NOTES:

i) IF $A_{p\times p} \geq 0$, then the spectral value decomposition = singular value decomposition.

HOWEVER, THIS IS NOT TRUE FOR SYMMETRIC A IN GENERAL. (WHY?)

ii) FOR $M_{p\times r}$, $A = MM'$ and $B = M'M$ HAVE THE SAME NON-ZERO EIGENVALUES, say $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_{MIN(P,r)}$, with the singular values of M being $\delta_j = \sqrt{\lambda_j}$, $j = 1, \ldots, MIN(P,r)$.

ALSO, $Q = (\underset{\sim}{q}_1, \ldots, \underset{\sim}{q}_p)$ ARE the eigenvectors of A, $A\underset{\sim}{q}_j = \lambda_j \underset{\sim}{q}_j$

and $H = (\underset{\sim}{h}_1, \ldots, \underset{\sim}{h}_r)$ ARE the eigenvectors of B, $B\underset{\sim}{h}_j = \lambda_j \underset{\sim}{h}_j$

FURTHER MORE $A\underset{\sim}{N} = \lambda_j \underset{\sim}{N}$ IFF $B(M'\underset{\sim}{N}) = \lambda_j (M'\underset{\sim}{N})$

HOMEWORK SET 3 TOPICS UP TO THIS POINT

NEXT TOPIC: MULTIVARIATE NORMAL DISTRIBUTION